Theoretical Computer Science 957 (2023) 113824

Contents lists available at ScienceDirect

Theoretical Computer Science

journal homepage: www.elsevier.com/locate/tcs

Secure connected domination and secure total domination in unit disk graphs and rectangle graphs ☆

Cai-Xia Wang, Yu Yang, Shou-Jun Xu *

School of Mathematics and Statistics, Gansu Center for Applied Mathematics, Lanzhou University, Lanzhou, Gansu 730000, China

A R T I C L E I N F O

Article history:
Received 15 December 2022
Received in revised form 18 February 2023
Accepted 12 March 2023
Available online 23 March 2023
Communicated by D.-Z. Du

Keywords:
Secure connected domination
Secure total domination
Unit disk graphs
Rectangle graphs
NP-complete
Approximation algorithm

A B S T R A C T

Given a graph G with vertex set V, a secure connected (resp. total) dominating set of G is a connected (resp. total) dominating set $S \subseteq V$ with the property that for each $u \in V \setminus S$, there exists $v \in S$ adjacent to u such that $(S \cup \{u\}) \setminus \{v\}$ is a connected (resp. total) dominating set of G. The minimum secure connected dominating set (or, for short, MSCDS) (resp. minimum secure total dominating set (or, for short, MSTDS)) problem is to find an MSCDS (resp. MSTDS) in a given graph.

In this paper, we initiate to consider complexity and algorithmic aspects of the MSCDS problem and the MSTDS problem in unit disk graphs and rectangle graphs. Firstly, we show that the decision version of the MSCDS problem is NP-complete in unit square graphs and unit disk graphs. Then we show that the decision version of the MSTDS problem is NP-complete even in grid graphs (a subclass of unit square graphs and unit disk graphs). Secondly, we give linear-time constant-approximation algorithms for the two problems in unit square graphs, unit disk graphs and unit-height rectangle graphs. Thirdly, we propose a PTAS for the MSTDS problem in unit square graphs and unit disk graphs. Finally, we show that the two problems in proper rectangle graphs are APX-hard. Further we give an explicit lower bound 1.00147 on efficient approximability for the two problems in proper rectangle graphs unless P=NP.

1. Introduction

Let $G = (V, E)$ be a simple, undirected connected graph with vertex set V and edge set E. Denote the *open* and *closed neighborhood* of a vertex $v \in V$ by $N_G(v)$ and $N_G[v]$, respectively, that is, $N_G(v) = \{x \in V : vx \in E\}$ and $N_G[v] = N_G(v) \cup \{v\}$. For a set $S \subseteq V$, an S-*external private neighbor* of a vertex $v \in S$ is a vertex $u \in V \setminus S$ such that $N_G(u) \cap S = \{v\}$. The S-*external private neighborhood* of $v \in S$, denoted by $epn(v, S)$, is the set of all S-external private neighbors of v. The *degree* of a vertex in G is the cardinality of its open neighborhood. A *leaf* of graph G is a vertex of degree one and the unique neighbor of a leaf is called *support vertex*. We denote by $L(G)$ and $S(G)$ be the sets of leaves and support vertices of G, respectively. For $V' \subseteq V$, the *induced subgraph by* V', denoted by $G[V']$, is the graph with vertex set V', in which two vertices are adjacent if and only if they are adjacent in G.

A *vertex cover* of graph G is a set $C \subseteq V$ with $u \in C$ or $v \in C$ for any edge $uv \in E$. The *vertex cover number* $\beta(G)$ is the minimum cardinality among all vertex covers of G. An *independent set* of graph G is a set $I \subseteq V$ such that any two vertices

☆ This article belongs to Section A: Algorithms, automata, complexity and games, Edited by Paul Spirakis.

* Corresponding author.
E-mail address: shjxu@lzu.edu.cn (S.-J. Xu).

https://doi.org/10.1016/j.tcs.2023.113824

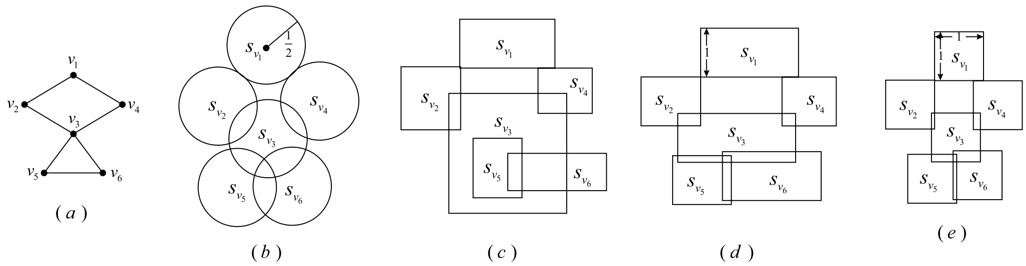

Fig. 1. (a) A graph G. (b) A unit disk representation of G. (c) A rectangle intersection representation of G. (d) A unit-height rectangle intersection representation of G. (e) A unit square intersection representation of G.

in I are not adjacent. An independent set I is referred as a *maximal independent set* if for any vertex $v \in V \setminus I$, $I \cup \{v\}$ is not an independent set. The *independence number* $\alpha(G)$ of G is the maximum cardinality among all independent sets of G. We say that a vertex $u \in V$ is *dominated* by a vertex $v \in V$ if $uv \in E$. A set $S \subseteq V$ is *a dominating set* (or, for short, *DS*) of G if each vertex not in S is dominated by at least one vertex of S. A DS S of G is said to be a *connected dominating set* (or, for short, *CDS*) (resp. *total dominating set* (or, for short, *TDS*)) of G, if the induced subgraph $G[S]$ is connected (resp. $G[S]$ has no isolated vertex).

In the context of graph protection, Cockayne et al. in [12] introduced the concept of secure domination. A *secure dominating set* of G is a DS S of G with the property that for each vertex $u \in V \setminus S$, there exists $v \in S$ adjacent to u such that $(S \cup \{u\}) \setminus \{v\}$ is a DS of G. The connected version and the total version of secure domination have also been proposed. A *secure connected dominating set* (or, for short, an *SCDS*) of G, introduced by Cabaro et al. in [8], is a CDS S of G with the property that for each $u \in V \setminus S$, there exists $v \in S$ adjacent to u such that $(S \cup \{u\}) \setminus \{v\}$ is a CDS of G. A *secure total dominating set* (or, for short, an *STDS*) of G, introduced by Benecke et al. in [4], is a TDS S of G with the property that for each $u \in V \setminus S$, there exists $v \in S$ adjacent to u such that $(S \cup \{u\}) \setminus \{v\}$ is a TDS of G. In these cases, we call v is an *S-defender* of u. The *secure connected domination number* (resp. *secure total domination number*) $\gamma_{sc}(G)$ (resp. $\gamma_{st}(G)$) is the minimum cardinality among all SCDSs (resp. STDSs) of G. Clearly, an SCDS is also an STDS, so $\gamma_{st}(G) \leq \gamma_{sc}(G)$. The definitions of SCDS and STDS imply the following results.

Theorem 1.1. *[8,25] Let G be a connected graph, S an SCDS or STDS of G. Then*
(i) $L(G) \cup S(G) \subseteq S$;
(ii) No vertex in $L(G) \cup S(G)$ is an S-defender.

Theorem 1.2. *[4,7] Let G be a connected graph, S an SCDS or STDS of G. Then $epn(v, S) = \emptyset$ for all $v \in S$.*

Lemma 1.3. *[22] Let G be a connected graph, C a CDS of G and D a DS of $G \setminus C$. Then $C \cup D$ is an SCDS of G.*

A *geometric intersection representation* of a graph $G = (V, E)$ is a set of geometric objects $P = \{s_v : v \in V\}$ such that each vertex $v \in V$ corresponds to a geometric object s_v and $uv \in E$ if and only if $s_u \cap s_v \neq \emptyset$. If the height and the width of a rectangle are k and l, respectively, then we say the rectangle has *size* $k \times l$, where k and l are positive real numbers. When P is a set of unit disks (the radius of all disks is $\frac{1}{2}$) (resp. axis-parallel rectangles, axis-parallel unit-height rectangles (the height and width of all rectangles are 1 and at least 1, respectively), axis-parallel unit squares (the size of all squares is 1×1)), P is called a *unit disk representation*, (resp, *rectangle intersection representation*, *unit-height rectangle intersection representation*, *unit square intersection representation*). If a graph has a geometric intersection representation, then the graph is called a *geometric intersection graph*. In particular, if a graph has a unit disk representation (resp. rectangle intersection representation, unit-height rectangle intersection representation, unit square intersection representation), then the graph is called a *unit disk graph* (resp. *rectangle graph, unit-height rectangle graph, unit square graph*), see Fig. 1. If centers of all disks of a unit disk graph have integer coordinates, then the graph is called a *grid graph*. Obviously, grid graphs are a subclass of unit disk graphs. A rectangle graph is *proper* if no rectangle is properly contained within another.

Among countless classes of geometric intersection graphs, unit disk graphs are extensively studied, owing their widely application in wireless ad hoc networks. Different from wired networks, wireless ad hoc networks have no physical backbone infrastructure. But a virtual backbone can be formed by a CDS. Unfortunately, computing a minimum CDS is NP-hard even in unit disk graphs. Therefore, a large number of approximation algorithms [10,15,19,24,28] have been designed for the minimum CDS in unit disk graphs. TDS has also been studied in unit disk graphs [20,31]. Another important class of geometric intersection graphs is rectangle graphs. Maximum-weight independent sets in such graphs are widely studied due to their application in data mining, map labeling, and VLSI design [1,2,9]. Wang et al. [29,30] studied the complexity and algorithm for the minimum secure domination problem in unit disk graphs and rectangle graphs.

The *MSCDS problem* (resp. *MSTDS problem*) is the problem of finding a minimum SCDS (resp. STDS) in a given graph. Given a positive integer j and a graph G, the *decision version* of the MSCDS (resp. MSTDS) problem is to decide whether G has an

SCDS (resp. STDS) of cardinality at most j. The decision version of the MSCDS problem was shown to be NP-complete for split graphs [23], chordal bipartite graphs, star convex bipartite graphs, comb convex bipartite graphs and doubly chordal graphs [22]. The decision version of the MSTDS problem was shown to be NP-complete for split graphs [13,23], chordal bipartite graphs, planar bipartite graphs with arbitrary large girth and maximum degree 3, graphs of separability at most 2 [13] and circle graphs [26]. The MSCDS problem is linear-time solvable in block graphs, threshold graphs [23] and chain graph [22]. The MSTDS problem is linear-time solvable in trees [26], cographs and chain graphs [21]. Kumar et al. [22] gave a $(\Delta(G)+1)$-approximation algorithm for the MSCDS problem. Duginov [13] showed that the MSTDS problem can be approximated in polynomial-time within a factor of $c\ln|V|$ for some constant $c>1$. The MSCDS problem [22] and the MSTDS problem [26] cannot be approximated within $(1-\varepsilon)\ln|V|$ for any $\varepsilon>0$, unless NP$\subseteq$DTIME($|V|^{O(\log\log|V|)}$) even for bipartite graphs and they are APX-complete for graphs with maximum degree 4.

In this paper, we consider complexity and algorithmic aspects of the MSCDS problem and the MSTDS problem in unit disk graphs and rectangle graphs. In section 2, we show that the decision version of the MSCDS problem is NP-complete in unit square graphs and unit disk graphs. Then we show that the decision version of the MSTDS problem is NP-complete even in grid graphs (a subclass of unit square graphs [5] and unit disk graphs). In section 3, we give linear-time constant-approximation algorithms for the two problems in unit square graphs, unit disk graphs and unit-height rectangle graphs. In section 4, we propose a PTAS for the MSTDS problem in unit square graphs and unit disk graphs. In section 5, we show that the two problems in proper rectangle graphs are APX-hard. Further we give an explicit lower bound 1.00147 on efficient approximability for the two problems in proper rectangle graphs unless P=NP.

2. NP-completeness

In this section, we consider complexity of the MSCDS problem and the MSTDS problem in unit square graphs and unit disk graphs. We first show that the decision version of the MSCDS problem is NP-complete in unit square graphs and unit disk graphs. Then we show that the decision version of the MSTDS problem is NP-complete in grid graphs (a subclass of unit square graphs [5] and unit disk graphs). The associated decision problems are defined below.

Secure Connected Domination (SCDM)
Instance: A graph G, $j\in\mathbb{N}^+$.
Question: Does G have an SCDS of size at most j?

Secure Total Domination (STDM)
Instance: A graph G, $j\in\mathbb{N}^+$.
Question: Does G have an STDS of size at most j?

Domination in Planar graphs (DM-PLA)
Instance: A planar graph G with maximum degree 3, $k\in\mathbb{N}^+$.
Question: Does G have a DS of size at most k?

Vertex Cover in Planar graphs (VC-PLA)
Instance: A planar graph G with maximum degree 3, $k\in\mathbb{N}^+$.
Question: Does G have a vertex cover of size at most k?

Let G be a planar graph. An *orthogonal drawing* of G is an embedding of G in the plane in such a way that its vertices are located at points with integer coordinates and its edges are drawn as a sequence of connected line segments of the form $x=i$ (vertical) or $y=j$ (horizontal) for integers i and j. A graph G admits an orthogonal drawing if and only if it is planar with maximum degree at most 4 [27]. The following lemma summarizes and strengthens the statement.

Lemma 2.1. *[6] For a planar graph G with vertex set V and maximum degree at most 4, there is a linear-time algorithm that produces an orthogonal drawing of G with, along each edge e, at most 2 bends (i.e., a point of e where a horizontal and a vertical segment meet) and using* $O(|V|^2)$ *area.*

Firstly, we show that SCDM is NP-complete in unit square graphs by proposing a polynomial-time reduction from the NP-complete problem DM-PLA [17].

Theorem 2.2. *SCDM is NP-complete in unit square graphs.*

Proof. For a given graph $G=(V,E)$, a set $S\subseteq V$ and $j\in\mathbb{N}^+$, we can verify whether S is an SCDS of G with $|S|\le j$ or not in polynomial-time. So SCDM is in NP. In what follows we offer a polynomial-time reduction from the NP-complete problem DM-PLA [17]. Let $G=(V,E)$ be a connected planar graph with maximum degree 3, where $V=\{v_1,v_2,\cdots,v_n\}$ and $|E|=m$. We construct a unit square graph $G'=(V',E')$ from G as follows:

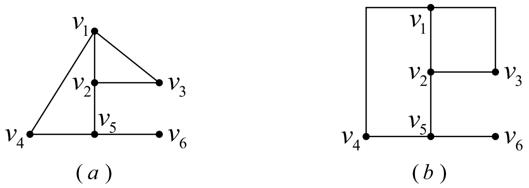

Fig. 2. (a) A connected planar graph G with maximum degree 3. (b) An orthogonal drawing of G.

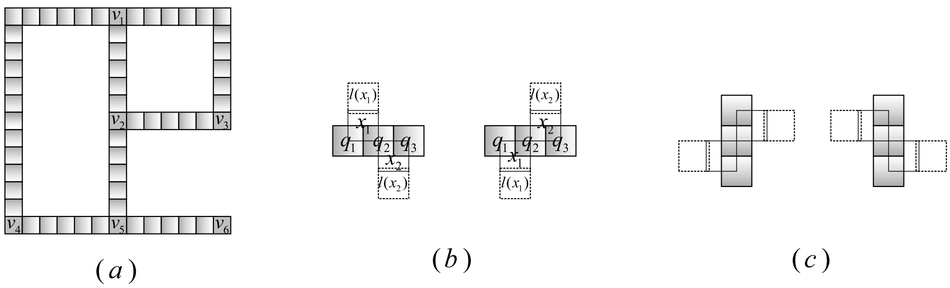

Fig. 3. (a) The drawing Γ'. (b) Three horizontal consecutive squares in Γ'. $x_1 = (a-0.5, b+0.5)$, $l(x_1) = (a-0.5, b+1.4)$, $x_2 = (a+0.5, b-0.5)$ and $l(x_2) = (a+0.5, b-1.4)$ (the left figure), or $x_1 = (a-0.5, b-0.5)$, $l(x_1) = (a-0.5, b-1.4)$, $x_2 = (a+0.5, b+0.5)$ and $l(x_2) = (a+0.5, b+1.4)$ (the right figure). (c) Three vertical consecutive squares in Γ'.

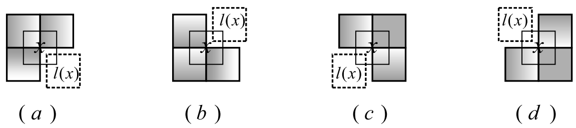

Fig. 4. (a) $l(x) = (a+0.7, b-0.7)$. (b) $l(x) = (a+0.7, b+0.7)$. (c) $l(x) = (a-0.7, b-0.7)$. (d) $l(x) = (a-0.7, b+0.7)$.

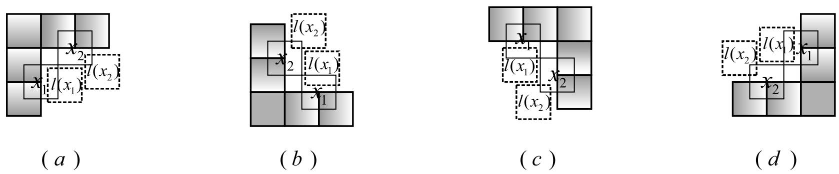

Fig. 5. (a) $l(x_1) = (a+0.7, b-0.1)$, $l(x_2) = (a+1.8, b+0.3)$. (b) $l(x_1) = (a+0.1, b+0.7)$, $l(x_2) = (a-0.3, b+1.8)$. (c) $l(x_1) = (a-0.1, b-0.7)$, $l(x_2) = (a+0.3, b-1.8)$. (d) $l(x_1) = (a-0.7, b+0.1)$, $l(x_2) = (a-1.8, b-0.3)$.

Step 1. Firstly, construct an orthogonal drawing of G by Lemma 2.1, see Fig. 2 (b). Secondly, expand the drawing by 6 times, denoted the resulted drawing by Γ, and for each vertex v_i ($1 \le i \le n$), label still as v_i the grid point at which v_i is located in Γ. Obviously, Γ is also an orthogonal drawing of G. Denote the sequence of connected line segment(s) in Γ representing the edge $v_i v_j$ in G by L_{ij}. Obviously, the length of the L_{ij} is divided by 6. We divide the sequence L_{ij} into l_{ij} part(s) with each being of length 6.
Step 2. Firstly, for each grid point (the point with integer coordinates) of Γ, add a unit square such that its center is located at the grid point, see Fig. 3 (a), denoted the resulted drawing by Γ'. In the following, we do not distinguish a square and its center. There are $6l_{ij} - 1$ squares on L_{ij} except the squares v_i, v_j for each $v_i v_j \in E$. Secondly, for each three horizontal consecutive squares q_1, q_2, q_3 in Γ', that is, $q_1 = (a-1, b)$, $q_2 = (a, b)$, $q_3 = (a+1, b)$, add four unit squares x_1, x_2 and $l(x_1), l(x_2)$ such that: (1) x_1 intersects with q_1 and q_2, x_2 intersects with q_2 and q_3, (2) x_1 and x_2 intersect at a corner, (3) $l(x_i)$ only intersects with x_i, where $i = 1, 2$. There are two cases, the details see Fig. 3 (b). We call x_1, x_2 are *connected squares*, $l(x_1), l(x_2)$ are *leaf squares*. The case of three vertical consecutive squares is similar, see Fig. 3 (c). Thus, each connected square x corresponds a leaf square $l(x)$. Let C (resp. L) be the set of all connected (resp. leaf) squares. Note that when one of the q_1, q_2, q_3 is a bend of Γ, the placement of the leaf square given in Fig. 3 (b) and (c) may cause a confusion. The following step is used to handle the confusion.
Step 3. At a bend of Γ, there are two cases for the placement of connected squares. The first case, a connected square $x = (a, b)$ intersects three squares in Γ' as shown in Fig. 4. Then move the leaf square $l(x)$, the details see Fig. 4. The second case, two connected squares x_1 and x_2 intersect two squares in Γ' respectively as shown in Fig. 5. Then move the leaf squares $l(x_1)$ and $l(x_2)$, the details see Fig. 5, where suppose $x_1 = (a, b)$. See Fig. 6 (a) for the drawing obtained from Step 2 and Step 3.

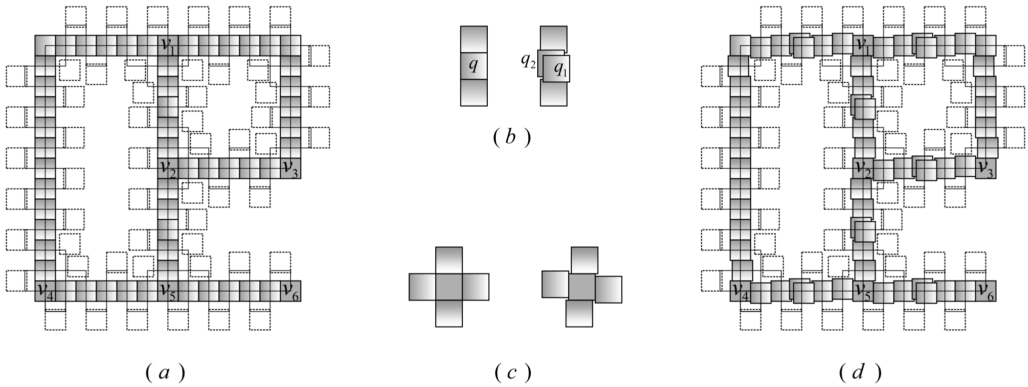

Fig. 6. (a) The drawing obtained from Step 2 and Step 3. (b) The replacement of squares for each L_{ij}. (c) Slight movement of squares. (d) A unit square intersection representation of a unit square graph G' obtained from G given in Fig. 2 (a). The shaded squares indicate the squares in $V \cup Q$, the blank squares with solid (resp. dashed) lines indicate the squares in C (resp. L).

Step 4. Firstly, for each L_{ij}, remove the fourth square $q = (a, b)$ staring from v_i or v_j but not both (except the case $l_{ij} = 1$). Then add two unit squares $q_1 = (a + 0.1, b - 0.1), q_2 = (a - 0.1, b + 0.1)$, see Fig. 6 (b). Thus there are $6l_{ij}$ squares on L_{ij} except the squares v_i, v_j for each $v_iv_j \in E$, denoted the sequence of the squares on L_{ij} starting from v_i by $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$. Secondly, for each square whose center is a bend of Γ, we move up (resp. down, right, left) the square on the left (resp. right, upper, lower) by 0.1, see Fig. 6 (c).

Refer Fig. 6 (d) is a unit square intersection representation of a unit square graph G' obtained from the graph G given in Fig. 2 (a). Let $Q = \cup_{v_iv_j \in E}\{q_1, \cdots, q_{6l_{ij}}\}$. Then $V' = V \cup Q \cup C \cup L$. Clearly, $|C| = |L| = O(l)$, where $l = \sum_{v_iv_j \in E} l_{ij}$. By Lemma 2.1, $l = O(n^2)$, it follows that $|V'| = n + 2|C| + 6l = O(n^2)$. Then G' can be constructed in polynomial-time. Note that each square in L is a leaf in G'. In what follows we show that G has a DS of size at most k if and only if G' has an SCDS of size at most $k + 2|C| + 2l$.

Let D be a DS of G with size at most k. We will construct an SCDS S of G' with $|S| = |D| + 2|C| + 2l$. Firstly, add $D \cup C \cup L$ into S. Secondly, for each edge v_iv_j in G, we consider the sequence $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$. If $v_i \in D$ or $v_j \in D$, without loss of generality, suppose $v_i \in D$, then add q_{3t} into S for all $1 \le t \le 2l_{ij}$. Otherwise $v_i, v_j \notin D$, then add q_{3t-1} into S for all $1 \le t \le 2l_{ij}$. Clearly, $|S| = |D| + 2|C| + 2l$.

Now we show that S is an SCDS of G'. By the construction of G', $C \cup L$ is a CDS of G'. By Lemma 1.3, we only need to show that $S \cap (V \cup Q)$ is a DS of $G' \setminus (C \cup L) = G'[V \cup Q]$. For each sequence $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$, if $v_i, v_j \notin S$, then q_{3t-2} and q_{3t} are dominated by q_{3t-1} for all $1 \le t \le 2l_{ij}$. If $v_i \in S$ or $v_j \in S$, without loss of generality, suppose $v_i \in S$, then q_1 is dominated by v_i, q_{3t-1} and q_{3t+1} are dominated by q_{3t} for all $1 \le t \le 2l_{ij} - 1$, and $q_{6l_{ij}-1}$ is dominated by $q_{6l_{ij}}$. For each $v_j \notin S$, then $v_j \notin D$. There is $v_t \in D$ such that $v_tv_j \in E$. Then v_j is dominated by $q_{6l_{tj}} \in S$ by the construction of S, where $q_{6l_{tj}}$ is in the sequence $(v_t, q_1, \cdots, q_{6l_{tj}}, v_j)$. Thus $S \cap (V \cup Q)$ is a DS of $G'[V \cup Q]$. Hence S is an SCDS of G'.

Let S be an SCDS of G' with size at most $k + 2|C| + 2l$ such that $|S \cap Q|$ is as small as possible. By Theorem 1.1, $C \cup L \subseteq S$ and each vertex in $C \cup L$ is not an S-defender. For each $p \in (V \cup Q) \setminus S$, there is an S-defender $p' \in S \cap (V \cup Q)$ of p, it means that each vertex in $(V \cup Q) \setminus S$ has a neighbor in $S \cap (V \cup Q)$. Thus $D_1 = S \cap (V \cup Q)$ is a DS of $G'[V \cup Q]$.

Claim 2.3. *For each sequence $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$ and each $1 \le t \le 2l_{ij}$, $|D_1 \cap \{q_{3t-2}, q_{3t-1}, q_{3t}\}| = 1$.*

Proof. Clearly for each sequence $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$ and each $1 \le t \le 2l_{ij}$, $|D_1 \cap \{q_{3t-2}, q_{3t-1}, q_{3t}\}| \ge 1$. Suppose to the contrary that there is a sequence $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$ such that $|D_1 \cap \{q_{3t-2}, q_{3t-1}, q_{3t}\}| \ge 2$ for some $1 \le t \le 2l_{ij}$. If $v_i, v_j \notin D_1$, then define $D_2 = (D_1 \setminus \{q_1, \cdots, q_{6l_{ij}}\}) \cup (\cup_{t=1}^{2l_{ij}}\{q_{3t}\}) \cup \{v_i\}$. Otherwise $v_i \in D_1$ or $v_j \in D_1$, without loss of generality, suppose $v_i \in D_1$. Then define $D_2 = (D_1 \setminus \{q_1, \cdots, q_{6l_{ij}}\}) \cup (\cup_{t=1}^{2l_{ij}}\{q_{3t}\})$. In either case, D_2 is a DS of $G'[V \cup Q]$ with $|D_2| \le |D_1|$, but $|D_2 \cap Q| < |D_1 \cap Q|$. Note that $C \cup L$ is a CDS of G'. By Lemma 1.3, $S' = D_2 \cup C \cup L$ is an SCDS of G' with $|S'| \le |S|$, but $|S' \cap Q| < |S \cap Q|$, a contradiction. $\square$

Let $D = D_1 \cap V$. Now we show that D is a DS of G. For each $v_j \in V \setminus D$, since $D = D_1 \cap V$, $v_j \notin D_1$. Then there is $q_{6l_{ij}} \in D_1$ such that v_j is dominated by $q_{6l_{ij}}$ in $G'[V \cup Q]$, where $q_{6l_{ij}}$ is in a sequence $(v_i, q_1, \cdots, q_{6l_{ij}}, v_j)$, it implies $v_iv_j \in E$. By Claim 2.3, $q_{3t} \in D_1$ and $q_{3t-2}, q_{3t-1} \notin D_1$ for each $1 \le t \le 2l_{ij}$. Then $v_i \in D_1$ to dominate q_1. Thus $v_i \in D$, it follows that D is a DS of G. By Claim 2.3, $|S \cap Q| = \sum_{v_iv_j \in E}(2l_{ij}) = 2l$. Then $k + 2|C| + 2l \ge |S| = |D| + |C| + |L| + |S \cap Q| = |D| + 2|C| + 2l$, it follows that $|D| \le k$. $\square$

In what follows, we show that SCDM is NP-complete in unit disk graphs using a method similar to Theorem 2.2.

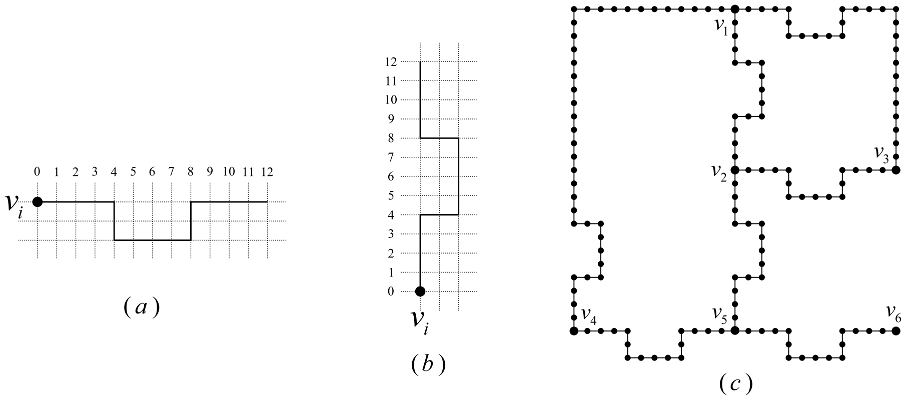

Fig. 7. (a) P is horizontal. (b) P is vertical. (c) A drawing Γ''.

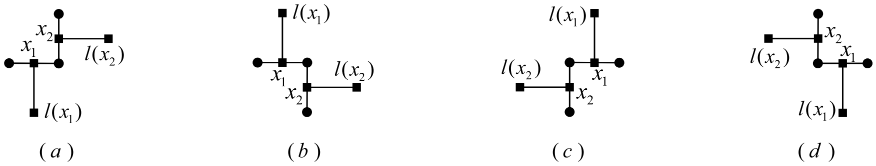

Fig. 8. (a) $l(x_1) = (a, b - \lambda_1)$, $l(x_2) = (c + \lambda_2, d)$. (b) $l(x_1) = (a, b + \lambda_1)$, $l(x_2) = (c + \lambda_2, d)$. (c) $l(x_1) = (a, b + \lambda_1)$, $l(x_2) = (c - \lambda_2, d)$. (d) $l(x_1) = (a, b - \lambda_1)$, $l(x_2) = (c - \lambda_2, d)$.

Theorem 2.4. *SCDM is NP-complete in unit disk graphs.*

Proof. By Theorem 2.2, SCDM is in NP. Similar to Theorem 2.2, in the following we offer a polynomial-time reduction from the NP-complete problem DM-PLA [17]. Given a simple connected planar graph $G = (V, E)$ with maximum degree 3, where $V = \{v_1, v_2, \cdots, v_n\}$ and $|E| = m$. We construct a unit disk graph $G' = (V', E')$ from G as follows:
Step 1. The construction of this step is the same as Step 1 of Theorem 2.2, but replace 6 with 12.
Step 2. Firstly, for each L_{ij}, replace the part P containing v_i or v_j but not both (except the case $l_{ij} = 1$) with the shape as presented in Fig. 7 (a) if P is horizontal, otherwise with the shape as presented in Fig. 7 (b), denoted the resulted drawing by Γ'. Secondly, add a vertex at each grid point of Γ', see Fig. 7 (c), denoted the resulted drawing by Γ''. Let Q be the set of all added vertices except the vertex in V. Then $|Q| = \sum_{v_i v_j \in E}(12l_{ij} + 3)$. Call each line segment of length 1 in Γ'' as *unit-line segment*. Note that each edge $v_i v_j$ in G corresponds a sequence of $12l_{ij} + 4$ connected unit-line segments in Γ''. Thirdly, add a vertex on the center point of each unit-line segment, and call these vertices are *connected vertices*, see Fig. 10.
Step 3. Each vertex in Γ'' connects at most three unit-line segments. For each connected vertex x, add a *leaf vertex* $l(x)$ as follows:
(1) For each vertex in Γ'' connecting two unit-line segments with one is horizontal and another is vertical, let $x_1 = (a, b)$ (resp. $x_2 = (c, d)$) be the connected vertex on the horizontal (resp. vertical) unit-line segment, where $|a - c| = 0.5$ and $|b - d| = 0.5$. Let $\lambda_1, \lambda_2 \in \{0.9, 1\}$. There are four cases, Fig. 8 shows the placement of the leaf vertices $l(x_1)$ and $l(x_2)$ in each case.
(2) For each vertex in Γ'' connecting three unit-line segments, let x_1, x_2, x_3 be the three connected vertices on the three unit-line segments, respectively. Suppose x_1 and x_2 are on the same line, then there are four cases presented in Fig. 9. Let x_4 be the connected vertex lying on the same line with x_3 as shown in Fig. 9. We only discuss the case presented in Fig. 9 (a), the other three cases are similar. Suppose $x_i = (a_i, b_i)$, where $i = 1, 2, 3, 4$. First move x_3 to $(a_3 + 0.5, b_3 - 0.5)$. Then add leaf vertices $l(x_1) = (a_1, b_1 + \lambda_1)$, $l(x_2) = (a_2, b_2 + \lambda_2)$, $l(x_3) = (a_3 + 1.1, b_3 - 0.6)$ and $l(x_4) = (a_4 - \lambda_4, b_4)$, where $\lambda_1, \lambda_2, \lambda_4 \in \{0.9, 1\}$ and $\lambda_1 \neq \lambda_2$.
See Fig. 10 for the drawing obtained after (1) and (2), and denoted by Λ.
(3) Let $(x_0, x_1, \cdots, x_s, x_{s+1})$ be a sequence of connected vertices which are lying on a line segment in Λ with endpoints x_0 and x_{s+1}, and are satisfying $l(x_0), l(x_{s+1})$ have been added in (1) or (2), and $l(x_i)$ has not been added for any $1 \leq i \leq s$. We only discuss the case of the line segment is horizontal, the case of the line segment is vertical is similar. Suppose $x_i = (a_i, b)$ for any $0 \leq i \leq s+1$. Let $\lambda_1, \lambda_2 \in \{0.9, 1\}$ and $\lambda_1 \neq \lambda_2$. If $l(x_0)$ and $l(x_{s+1})$ are above the line segment, then $l(x_i) = (a_i, b + \lambda_1)$ and $l(x_{i+1}) = (a_{i+1}, b + \lambda_2)$ for any $0 \leq i \leq s$, see Fig. 11 (a). The case of $l(x_0)$ and $l(x_{s+1})$ are below the line segment is similar, see Fig. 11 (b). If $l(x_0)$ is above the line segment and $l(x_{s+1})$ is below the line segment, then $l(x_i) = (a_i, b + \lambda_1)$ and $l(x_{i+1}) = (a_{i+1}, b + \lambda_2)$ for any $0 \leq i \leq \lceil s/2 \rceil - 1$, and $l(x_i) = (a_i, b - \lambda_1)$ and $l(x_{i+1}) = (a_{i+1}, b - \lambda_2)$ for any $\lceil s/2 \rceil + 1 \leq i \leq s$, see Fig. 11 (c). The case of $l(x_0)$ is below the line segment and $l(x_{s+1})$ is above the line segment is similar, see Fig. 11 (d).

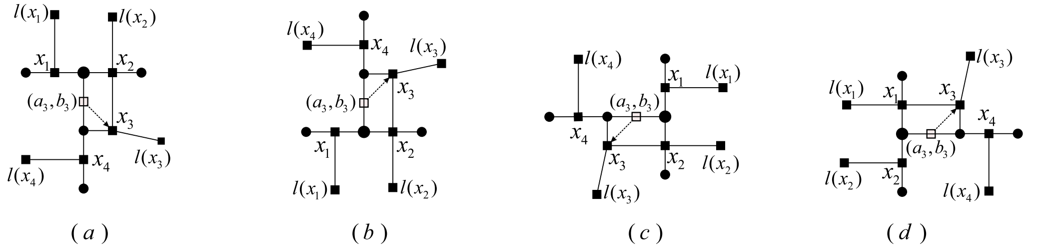

Fig. 9. A vertex in Γ'' connecting three unit-line segments.

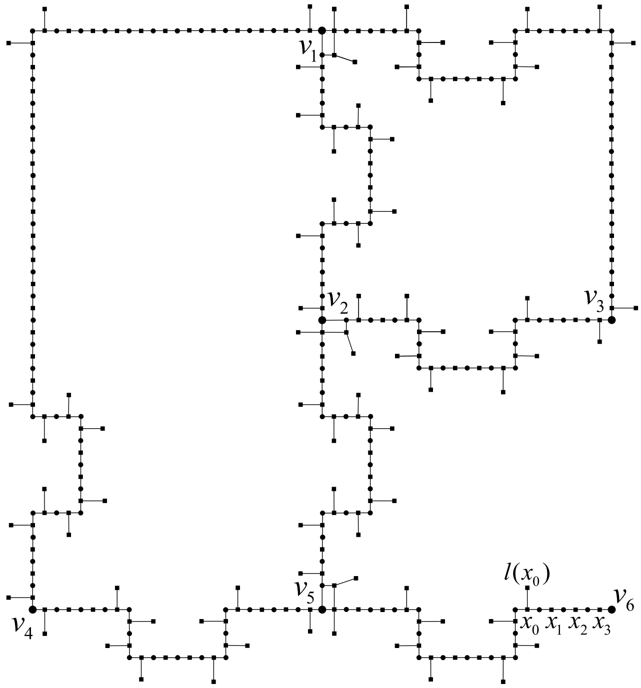

Fig. 10. The drawing Λ.

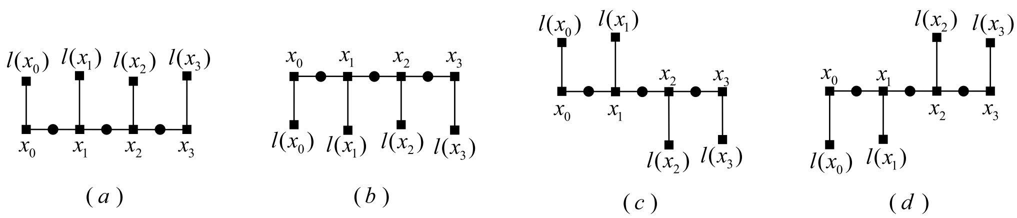

Fig. 11. An example of (3) when the line segment is horizontal and $s = 2$.

The remaining case is $l(x_{s+1})$ has not been added in (1) or (2) and x_{s+1} is distance 0.5 from a leaf of G. The case is similar with the first two cases. See Fig. 10 for an example, where $l(x_0)$ has been added in (1) or (2), $l(x_3)$ has not been added in (1) or (2) and x_3 is distance 0.5 from a leaf v_6 of G. Use the method in Fig. 11 (a) to add leaves $l(x_1), l(x_2), l(x_3)$. We denote the resulted drawing by Λ'.

Step 4. Let C be the set of connected vertices and L be the set of leaf vertices. Then $|C| = |L| = \sum_{v_i v_j \in E}(12l_{ij} + 4) = 12l + 4m$, where $l = \sum_{v_i v_j \in E} l_{ij}$. Now we construct the vertex set of G' as $V' = V \cup Q \cup C \cup L$, and two vertices in V' are adjacent if and only if they are of distance at most 1 in Λ', see Fig. 12. Note that each vertex in L is a leaf in G'.

Clearly, G' is a unit disk graph. Since the maximum degree of G is 3, $m = O(n)$. From Lemma 2.1, $l = O(n^2)$. Thus $|V'| = |V| + |Q| + |C| + |L| = n + 36l + 11m = O(n^2)$, it follows that G' can be constructed in polynomial-time. Use a method similar to Theorem 2.2, we can show that G has a DS of size at most k if and only if G' has an SCDS of size at most $k + 28l + 9m$. $\square$

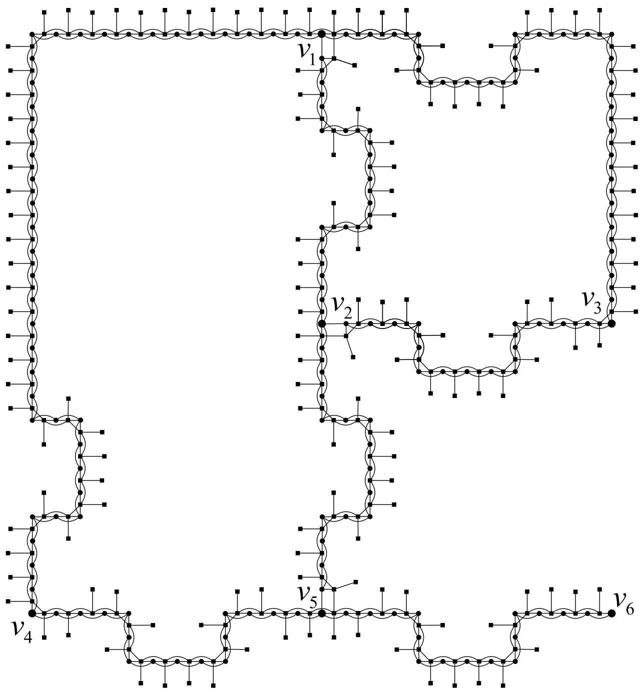

Fig. 12. A unit disk graph G' obtained from G given in Fig. 2 (a). The bigger dots (resp, smaller dots, squares) indicate the vertices in V (resp. Q, $C \cup L$).

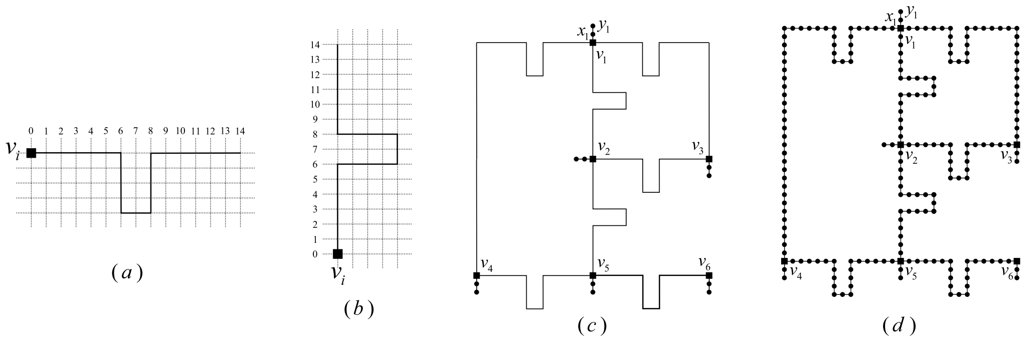

Fig. 13. (a) P is horizontal. (b) P is vertical. (c) A drawing Γ''. (d) Grid graph G'' obtained from G presented in Fig. 2 (a).

Finally, we show that STDM is NP-complete for grid graphs by proposing a polynomial-time reduction from the NP-complete problem VC-PLA [17].

Theorem 2.5. *STDM is NP-complete in grid graphs.*

Proof. Similar to Theorem 2.2, STDM is in NP. In what follows we offer a polynomial-time reduction from the NP-complete problem VC-PLA [17]. Let $G = (V, E)$ be a simple planar graph with maximum degree 3, where $V = \{v_1, v_2, \cdots, v_n\}$ and $|E| = m$. We use the same way as in [29] to construct a grid graph $G'' = (V'', E'')$ from G. In the following we redescribe the construction.

Step 1. The construction of this step is the same as Step 1 of Theorem 2.2, but replace 6 with 14.

Step 2. For each L_{ij}, replace the part P containing v_i or v_j but not both (except the case $l_{ij} = 1$) with the shape as presented in Fig. 13 (a) if P is horizontal, otherwise with the shape as presented in Fig. 13 (b). Denote the resulted sequence still by L_{ij} and the resulted drawing by Γ'. Note that the length of L_{ij} is $14l_{ij} + 8$. Since maximum degree of G is 3, there is a vertical or horizontal direction around each grid point labeled as v_i which is blank in Γ' and, further on the direction, we add a line segment S_i of length two with an endpoint v_i and label the grid points as x_i, y_i in turn (see example x_1, y_1 in Fig. 13 (c)), denoted the resulted drawing by Γ''.

Step 3. Now we construct the vertex set V'' of G'' as the grid points in Γ'', and two vertices are adjacent if and only if their corresponding points in Γ'' are of distance 1 (see Fig. 13 (d)).

Clearly, G'' is a grid graph. For each edge v_iv_j in G, there are $14l_{ij}+7$ vertices on L_{ij} except v_i, v_j, denoted $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$ by the sequence of the vertices on L_{ij} starting from v_i and $Q_t = \{q_{7t-5}, q_{7t-4}, q_{7t-2}, q_{7t-1}, q_{7t}\}$ for any $1 \le t \le 2l_{ij}+1$. Since the maximum degree of G is 3, $m = O(n)$. From Lemma 2.1, $l = \sum_{v_iv_j \in E} l_{ij} = O(n^2)$. Thus $|V''| = 3n + 14l + 7m = O(n^2)$, it follows that G'' can be constructed in polynomial-time. In the following we show that G has a vertex cover of size at most k if and only if G'' has an STDS of size at most $k + 2n + 10l + 5m$.

Let C be a vertex cover of G with size at most k. We will construct an STDS S of G'' with $|S| = |C| + 2n + 10l + 5m$. Firstly, add C and all x_i, y_i for $1 \le i \le n$ into S. Secondly, for each edge v_iv_j in G, $v_i \in C$ or $v_j \in C$, without loss of generality, assume $v_i \in C$. Then add Q_t for all $1 \le t \le 2l_{ij}+1$ into S. Clearly, $|S| = |C| + 2n + 10l + 5m \le k + 2n + 10l + 5m$. Now we show that S is an STDS of G''. Clearly, S is a TDS of G''. For each edge v_iv_j in G, we consider the sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$. Also we can assume $v_i \in C$, it follows that $v_i \in S$. By the construction of S, v_i is an S-defender of q_1, q_{7t-2} is an S-defender of q_{7t-3} for any $1 \le t \le 2l_{ij}+1$, q_{7t} is an S-defender of q_{7t+1} for any $1 \le t \le 2l_{ij}$, $q_{14l_{ij}+7}$ is an S-defender of v_j if $v_j \notin S$.

Let S be an STDS of G'' with size at most $k + 2n + 10l + 5m$ such that $|S \cap Q|$ is as small as possible, where $Q = \cup_{v_iv_j \in E}\{q_1, \cdots, q_{14l_{ij}+7}\}$. By Theorem 1.1, $x_i, y_i \in S$ and x_i, y_i are not S-defenders for any $1 \le i \le n$.

Claim 2.6. *For each sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$ and each $1 \le t \le 2l_{ij}+1$, $|\{q_{7t-6}, \cdots, q_{7t}\} \cap S| = 5$.*

Proof. We consider a sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$. For any $1 \le t \le 2l_{ij}+1$, if there are $q_{7t-s}, q_{7t-(s-1)} \notin S$, then $q_{7t-s}, q_{7t-(s-1)}$ have no S-defenders, where $1 \le s \le 6$, a contradiction. Thus for any $1 \le t \le 2l_{ij}+1$ and any $1 \le s \le 6$, $|\{q_{7t-s}, q_{7t-(s-1)}\} \cap S| \ge 1$. Then $|\{q_{7t-6}, \cdots, q_{7t}\} \cap S| \ge 4$ for any $1 \le t \le 2l_{ij}+1$. If there is $1 \le t \le 2l_{ij}+1$ such that $|\{q_{7t-6}, \cdots, q_{7t}\} \cap S| = 4$, then $q_{7t-5}, q_{7t-4}, q_{7t-2}, q_{7t-1} \in S$, it follows that q_{7t-3} has no S-defender, a contradiction. Thus for each sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$ and each $1 \le t \le 2l_{ij}+1$, $|\{q_{7t-6}, \cdots, q_{7t}\} \cap S| \ge 5$.

Now we show that for each sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$ and each $1 \le t \le 2l_{ij}+1$, $|\{q_{7t-6}, \cdots, q_{7t}\} \cap S| \le 5$. Suppose to the contrary that there is a sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$ such that $|\{q_{7t-6}, \cdots, q_{7t}\} \cap S| > 5$ for some $1 \le t \le 2l_{ij}+1$. If $v_i \notin S$ or $v_j \notin S$, without loss of generality, assume $v_i \notin S$, then construct $S' = (S \setminus \{q_1, \cdots, q_{14l_{ij}+7}\}) \cup (\cup_{t=1}^{2l_{ij}+1} Q_t) \cup \{v_i\}$. Otherwise $v_i, v_j \in S$, then construct $S' = (S \setminus \{q_1, \cdots, q_{14l_{ij}+7}\}) \cup (\cup_{t=1}^{2l_{ij}+1}\{q_{7t-6}, q_{7t-5}, q_{7t-3}, q_{7t-2}, q_{7t-1}\})$. In either case, S' is an STDS of G'' with $|S'| \le |S|$, but $|\{q_1, \cdots, q_{14l_{ij}+7}\} \cap S'| < |\{q_1, \cdots, q_{14l_{ij}+7}\} \cap S|$, a contradiction with $|S \cap Q|$ is as small as possible. $\square$

Next we show that $C = V \cap S$ is a vertex cover of G. For each edge v_iv_j in G, if $v_i, v_j \in C$, then it holds. So we assume $v_j \notin C$ or $v_i \notin C$, without loss of generality, suppose $v_j \notin C$, it follows that $v_j \notin S$. We consider the sequence $(v_i, q_1, \cdots, q_{14l_{ij}+7}, v_j)$. Then $q_{14l_{ij}+7}, q_{14l_{ij}+6} \in S$, otherwise $q_{14l_{ij}+7}$ has no S-defender. Combined with Claim 2.6, either $\{q_{7t-6}, \cdots, q_{7t}\} \cap S = Q_t$ or $\{q_{7t-6}, \cdots, q_{7t}\} \cap S = \{q_{7t-5}, q_{7t-4}, q_{7t-3}, q_{7t-1}, q_{7t}\}$ for any $1 \le t \le 2l_{ij}+1$. In either case, $q_1 \notin S$, then $v_i \in S$, otherwise q_1 has no S-defender. Then $v_i \in C$. Thus C is a vertex cover of G. Since $|S| = |C| + 2n + |S \cap Q| \le k + 2n + 10l + 5m$ and $|S \cap Q| = 10l + 5m$ by Claim 2.6, $|C| \le k$. $\square$

Grid graphs are a subclass of unit square graphs [5] and unit disk graphs. By Theorems 2.2 and 2.5, we have the following results.

Corollary 2.7. *STDM is NP-complete in unit square graphs and unit disk graphs. SCDM and STDM are NP-complete in unit-height rectangle graphs and rectangle graphs.*

3. Constant-approximation algorithm

In this section, we give linear-time constant-approximation algorithms for the MSCDS problem and the MSTDS problem in unit square graphs, unit disk graphs and unit-height rectangle graphs. We first present a lower bound of $\gamma_{sc}(G)$ and $\gamma_{st}(G)$ in terms of $\alpha(G)$ in a $K_{1,k+1}$-free graph G.

Lemma 3.1. *Let $G = (V, E)$ be a $K_{1,k+1}$-free connected graph, where $k \ge 1$ is an integer. Then $\alpha(G) \le \frac{k}{2}\gamma_{st}(G) \le \frac{k}{2}\gamma_{sc}(G)$.*

Proof. Clearly, $\gamma_{st}(G) \le \gamma_{sc}(G)$. Thus we only need to prove the first inequality. Let S be a minimum STDS of G, I a maximum independent set of G, $I_1 = I \cap S$ and $I_2 = I \cap (V \setminus S)$. Then $I = I_1 \cup I_2$.

Consider a bipartite graph H with bipartition $S \setminus I_1$ and I, and $E(H) = \{uv \in E : u \in S \setminus I_1, v \in I\}$. Since G is $K_{1,k+1}$-free, each vertex in $S \setminus I_1$ has at most k neighbors in I. Then

$$|E(H)| \le k(|S| - |I_1|). \tag{3.1}$$

Since S is a TDS of G and I_1 is independent, each vertex in I_1 has at least one neighbor in $S \setminus I_1$. By Theorem 1.2, $epn(v, S) = \emptyset$ for each $v \in S$. Then each vertex in $V \setminus S$ has at least 2 neighbors in S. Thus each vertex in I_2 has at least 2 neighbors in $S \setminus I_1$. Then

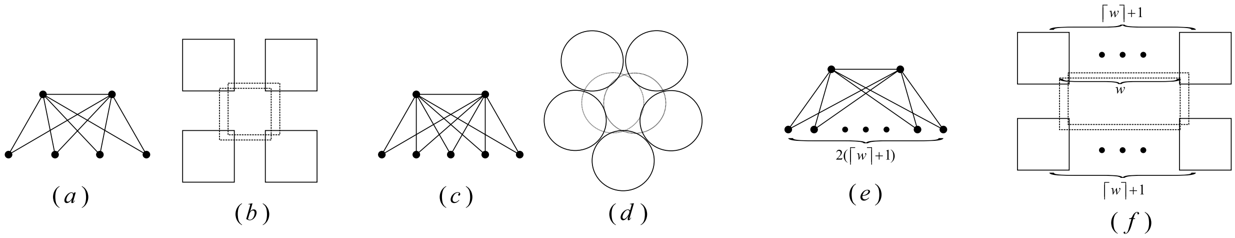

Fig. 14. (a) A unit square graph G with $\alpha(G) = 2\gamma_{st}(G) = 2\gamma_{sc}(G)$. (b) A unit square intersection representation of the graph presented in (a). (c) A unit disk graph G with $\alpha(G) = 2.5\gamma_{st}(G) = 2.5\gamma_{sc}(G)$. (d) A unit disk representation of the graph presented in (c). (e) A unit-height rectangle graph G with maximum width w and $\alpha(G) = (\lceil w \rceil + 1)\gamma_{st}(G) = (\lceil w \rceil + 1)\gamma_{sc}(G)$. (f) A unit-height rectangle intersection representation of the graph presented in (e).

$$|E(H)| \geq |I_1| + 2|I_2|. \tag{3.2}$$

Combining inequalities (3.1) with (3.2),

$$|I_2| \leq \frac{k|S| - (k+1)|I_1|}{2}. \tag{3.3}$$

By the inequality (3.3) and $k \geq 1$, we have the following

$$\begin{aligned}\alpha(G) &= |I_1| + |I_2| \\ &\leq |I_1| + \frac{k|S| - (k+1)|I_1|}{2} \\ &= \frac{k}{2}|S| - \frac{(k-1)}{2}|I_1| \\ &\leq \frac{k}{2}\gamma_{st}(G). \quad \square\end{aligned}$$

Lemma 3.2. *Given a unit-height rectangle intersection representation, then the corresponding unit-height rectangle graph G is $K_{1,2\lceil w \rceil+3}$-free, where $w \geq 1$ is the maximum width of all rectangles in the representation. Further, unit square graph is $K_{1,5}$-free.*

Proof. Let R be a unit-height rectangle with width w in the representation. Denote by B_t and B_b the top and bottom boundaries of R, respectively. Note that all unit-height rectangles intersecting with R are intersecting with B_t or B_b. There are at most $\lceil w \rceil + 1$ independent unit-height rectangles (that is, these rectangles do not intersect each other) intersecting with B_t. So as B_b. Then there are at most $2(\lceil w \rceil + 1)$ independent unit-height rectangles intersecting with R. Thus G is $K_{1,2\lceil w \rceil+3}$-free. $\square$

In the following we will give some examples to show that the bounds are sharp for unit square graphs, unit disk graphs and unit-height rectangle graphs. For a unit square graph G, $\alpha(G) \leq 2\gamma_{st}(G) \leq 2\gamma_{sc}(G)$ by Lemma 3.1 and Lemma 3.2. Fig. 14 (a) presents a unit square graph G with $\alpha(G) = 2\gamma_{st}(G) = 2\gamma_{sc}(G)$. For a unit disk graph G, $\alpha(G) \leq 2.5\gamma_{st}(G) \leq 2.5\gamma_{sc}(G)$ by Lemma 3.1 and the fact that unit disk graph is $K_{1,6}$-free [16]. Fig. 14 (c) presents a unit disk graph G with $\alpha(G) = 2.5\gamma_{st}(G) = 2.5\gamma_{sc}(G)$. For a unit-height rectangle graph whose unit-height rectangle intersection representation is given, $\alpha(G) \leq (\lceil w \rceil + 1)\gamma_{st}(G) \leq (\lceil w \rceil + 1)\gamma_{sc}(G)$, where $w \geq 1$ is the maximum width of all rectangles in the representation. Fig. 14 (e) presents a unit-height rectangle graph G with $\alpha(G) = (\lceil w \rceil + 1)\gamma_{st}(G) = (\lceil w \rceil + 1)\gamma_{sc}(G)$, where Fig. 14 (f) is the given unit-height rectangle intersection representation.

For a connected graph G, a CDS can be formed by a maximal independent set I and a set C of connectors with $|C| \leq |I| - 1$ in linear-time, see [28]. Thus by Lemma 1.3 and Lemma 3.1, we have the following result.

Lemma 3.3. *There is a linear-time $(1.5k)$-approximation algorithm for the MSCDS problem and the MSTDS problem in $K_{1,k+1}$-free connected graphs, where $k \geq 1$ is an integer.*

Proof. Given a $K_{1,k+1}$-free connected graph G. Let $I \cup C$ be a CDS of G produced in [28], where I is a maximal independent set of G and C is a set of connectors with $|C| \leq |I| - 1$. Let I' be a maximal independent set of $G \setminus (I \cup C)$. Then I' is a DS of $G \setminus (I \cup C)$. By Lemma 1.3, $I \cup C \cup I'$ is an SCDS of G, also is an STDS of G. Then by Lemma 3.1, $|I \cup C \cup I'| \leq 3\alpha(G) \leq (1.5k)\gamma_{st}(G) \leq (1.5k)\gamma_{sc}(G)$. $\square$

By Lemma 3.2, Lemma 3.3 and unit disk graph is $K_{1,6}$-free [16], we have the following results.

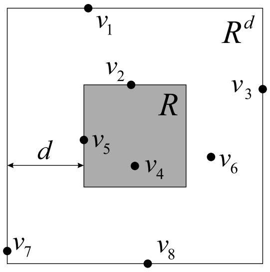

Fig. 15. The shaded square (resp. outmost square) indicates R (resp. R^d).

Theorem 3.4. *There is a linear-time 6-approximation (resp. 7.5-approximation) algorithm for the MSCDS problem and the MSTDS problem in connected unit square graphs (resp. unit disk graphs).*

Theorem 3.5. *There is a linear-time $3(\lceil w \rceil + 1)$-approximation algorithm for the MSCDS problem and the MSTDS problem in unit-height rectangle graphs whose unit-height rectangle intersection representation is given, where $w \geq 1$ is the maximum width of all rectangles in the representation.*

4. PTAS

In this section, we give a PTAS for the MSTDS problem in unit square graphs and unit disk graphs. We use the shifting strategy which was introduced by Hochbaum and Maass [18] to design the PTAS.

Given a set P of n unit squares such that the corresponding unit square graph has no isolated vertex. We first give some definitions. For a subset S of P, we say that $u \in S$ is *isolated* in S if there is no square in S intersecting with u. If there is no isolated square in S, then we say S is *total*, that is, for each $u \in S$, there is $v \in S$ such that $v \cap u \neq \emptyset$. Let Q, P' be subsets of P with $N[Q] \subseteq P'$, where $N[Q] = \{u \in P : u \in Q$ or there is $v \in Q$ such that $v \cap u \neq \emptyset\}$. A set $S \subseteq P'$ is a *dominating set of Q choosing from P'* (or, for short, *(Q, P')-DS*) if for each $u \in Q \setminus S$, there exists $v \in S$ such that $v \cap u \neq \emptyset$. A set $S \subseteq P'$ is a *total dominating set of Q choosing from P'* (or, for short, *(Q, P')-TDS*) if S is a (Q, P')-DS and S is total, in other words, for each $u \in Q \cup S$, there exists $v \in S$ such that $v \cap u \neq \emptyset$. A (Q, P')-TDS S is a *secure total dominating set of Q choosing from P'* (or, for short, *(Q, P')-STDS*) if for each $u \in Q \setminus S$, there is $v \in S$ with $v \cap u \neq \emptyset$ such that $(S \cup \{u\}) \setminus \{v\}$ is a (Q, P')-TDS. Note that $N[Q]$ is a (Q, P')-STDS, also a (Q, P')-TDS and a (Q, P')-DS, then the above three definitions are well defined. Let G be the unit square graph corresponding to P. Then a (P, P)-DS (resp. (P, P)-TDS, (P, P)-STDS) is a DS (resp. a TDS, an STDS) of G, which is also called a DS (resp. a TDS, an STDS) of P.

Now we discuss the shifting strategy. Let $\mathcal{R}$ be a smallest rectangular region containing centers of all squares in P and $k \geq 1$ be an integer. The shifting strategy contains 2 levels. The first level contains k iterations. In the ith iteration ($1 \leq i \leq k$), we use vertical lines to partition the region $\mathcal{R}$ into strips such that the width of the first strip is $10i$ and the remaining strips are $10k$ (the width of the last strip may less than $10k$). We consider any one of the strips with width $10k$ (the strip with width less than $10k$ can be considered similarly) and denote the strip by H. The second level contains k iterations. In the jth iteration ($1 \leq j \leq k$), we use horizontal lines to partition the strip H into rectangles such that the first rectangle has size $10j \times 10k$ and the remaining rectangles have size $10k \times 10k$ (the size of the last rectangle may less than $10k \times 10k$).

Let R be any one of the rectangles with size $10k \times 10k$ (the rectangles with size less than $10k \times 10k$ can be considered similarly). Expanding R by d units will obtain a rectangle R^d having the same center with R, where d is a nonnegative integer. Then R^d has size $(10k + 2d) \times (10k + 2d)$. Note that $R^0 = R$. Suppose the top and right boundaries of R^d are open except the boundaries of $\mathcal{R}$. Let $P_{R^d} \subseteq P$ be the set of squares whose centers are contained in R^d. See Fig. 15 for an example, where $v_1, \cdots, v_8$ are the centers of squares in P. Then $\{v_4, v_5\} \subseteq P_R$ and $\{v_2, v_4, v_5, v_6, v_7, v_8\} \subseteq P_{R^d}$. Define OPT_{R^2} as a minimum (P_{R^2}, P_{R^5})-STDS. Since $N[P_{R^2}] \subseteq P_{R^3} \subseteq P_{R^5}$, OPT_{R^2} is well defined.

Lemma 4.1. *OPT_{R^2} can be obtained in time $n_{R^5}^{O(k^2)}$, where $n_{R^5} = |P_{R^5}|$.*

Proof. Partition the rectangle R^2 into cells of size 1×1. There are $(10k + 4)^2$ cells in R^2. Suppose the top and right boundaries of each cell are open. If the center of a square is in a cell, then we say the square is in the cell. For each cell has only one square u, pick squares u and v, where v is a square intersecting with u (note that $v \in P_{R^3}$). Let $S^1_{R^2}$ be the set of all picked squares in this case. For each cell has at least one square not in $S^1_{R^2}$, if there is only one square in the cell not in $S^1_{R^2}$, then pick the square (note that in this case, the cell contains at least one square in $S^1_{R^2}$). Otherwise, pick any two squares not in $S^1_{R^2}$ from the cell. Let S_{R^2} be the set of all picked squares (including the squares in $S^1_{R^2}$). Then $|S_{R^2}| \leq 2(10k + 4)^2$.

Now we show that S_{R^2} is a (P_{R^2}, P_{R^5})-STDS. Clearly, $S_{R^2} \subseteq P_{R^3} \subseteq P_{R^5}$. Note that any two squares u and v in the same cell satisfy $u \cap v \neq \emptyset$. Then S_{R^2} is a (P_{R^2}, P_{R^5})-TDS. For any $u \in P_{R^2} \setminus S_{R^2}$, there are $v_1, v_2 \in S_{R^2} \setminus S^1_{R^2}$ such that

v_1, v_2, u are in the same cell. By the construction of S_{R^2}, we can also choose u, v_2 to put in S_{R^2} for the cell, it follows that $(S_{R^2} \cup \{u\}) \setminus \{v_1\}$ is a (P_{R^2}, P_{R^5})-TDS by the above discussion. Then S_{R^2} is a (P_{R^2}, P_{R^5})-STDS.

Since $|OPT_{R^2}| \le |S_{R^2}|$, OPT_{R^2} can be obtained by checking all subset in P_{R^5} with size at most $2(10k+4)^2$, which takes $O(n_{R^5}^{2(10k+4)^2}) = n_{R^5}^{O(k^2)}$ time. □

In the following we consider any (i, j)-th iteration, where $1 \le i, j \le k$. Suppose there are s rectangles $R_1, \cdots, R_s$ in this iteration. Let $S_{i,j} = OPT_{R_1^2} \cup \cdots \cup OPT_{R_s^2}$. The algorithm outputs a set S, which has minimum cardinality among all $\{S_{i,j} : 1 \le i, j \le k\}$, that is, $|S| = \min_{1\le i,j\le k}|S_{i,j}|$.

Lemma 4.2. *S is an STDS of P.*

Proof. We only need to prove $S_{i,j}$ is an STDS of P, for any $1 \le i, j \le k$. Since $P = P_{R_1^2} \cup \cdots \cup P_{R_s^2}$ and $OPT_{R_t^2}$ is a $(P_{R_t^2}, P_{R_t^5})$-TDS for any $1 \le t \le s$, $S_{i,j}$ is a TDS of P.

For any $u \in P \setminus S_{i,j}$, there is unique $1 \le t \le s$ such that $u \in P_{R_t}$. Then $u \in P_{R_t^2}$ and $u \notin OPT_{R_t^2}$ by $u \notin S_{i,j}$. Since $OPT_{R_t^2}$ is a $(P_{R_t^2}, P_{R_t^5})$-STDS, there is $v \in OPT_{R_t^2}$ with $v \cap u \ne \emptyset$ such that $(OPT_{R_t^2} \cup \{u\}) \setminus \{v\}$ is a $(P_{R_t^2}, P_{R_t^5})$-TDS. Now we show that $S'_{i,j} = (S_{i,j} \cup \{u\}) \setminus \{v\}$ is a TDS of P. If there is $x \in P$ such that x is not intersecting with any squares in $S'_{i,j} \setminus \{x\}$, then x is only intersecting with v in $S_{i,j}$ by $S_{i,j}$ is a TDS of P. Since $u \in P_{R_t}$, $v \in P_{R_t^1}$ by $v \cap u \ne \emptyset$, further $x \in P_{R_t^2}$ by $x \cap v \ne \emptyset$. Since $(OPT_{R_t^2} \cup \{u\}) \setminus \{v\}$ is a $(P_{R_t^2}, P_{R_t^5})$-TDS, there is $y \in (OPT_{R_t^2} \cup \{u\}) \setminus \{v\}$ such that $x \cap y \ne \emptyset$. Clearly, $y \in S'_{i,j}$, which is a contradiction with x is not intersecting with any squares in $S'_{i,j} \setminus \{x\}$. □

Lemma 4.3. *Let OPT be an STDS of P. Then $(OPT \cap P_{R_t^5}) \setminus I_{R_t^5}$ is a $(P_{R_t^2}, P_{R_t^5})$-STDS for any $1 \le t \le s$, where $I_{R_t^5} = \{p \in OPT \cap P_{R_t^5} :$ p is isolated in $OPT \cap P_{R_t^5}\}$.*

Proof. Let $S_{R_t^2} = (OPT \cap P_{R_t^5}) \setminus I_{R_t^5}$. Clearly, $S_{R_t^2} \subseteq P_{R_t^5}$. We first show that $I_{R_t^5} \cap P_{R_t^3} = \emptyset$. Suppose to the contrary that there is $p \in I_{R_t^5} \cap P_{R_t^3}$. Then $p \in OPT$. Since OPT is total, there is $q \in OPT$ such that $q \cap p \ne \emptyset$. Then $q \in P_{R_t^5}$ by $p \in P_{R_t^3}$, it follows that p is not isolated in $OPT \cap P_{R_t^5}$, which is a contradiction with $p \in I_{R_t^5}$. Thus

$$(OPT \cap P_{R_t^3}) \subseteq S_{R_t^2}. \tag{*}$$

Clearly, $S_{R_t^2}$ is total. For each $u \in P_{R_t^2} \setminus S_{R_t^2}$, $u \notin OPT$ (otherwise $u \in OPT \cap P_{R_t^2} \subseteq OPT \cap P_{R_t^3} \subseteq S_{R_t^2}$ by (*), a contradiction). Since OPT is an STDS of P, there is $v \in OPT$ with $v \cap u \ne \emptyset$ such that $(OPT \cup \{u\}) \setminus \{v\}$ is a TDS of P. Then $v \in P_{R_t^3}$ by $u \in P_{R_t^2}$, it follows that $v \in S_{R_t^2}$ by (*). Thus $S_{R_t^2}$ is a $(P_{R_t^2}, P_{R_t^5})$-TDS.

We now show that $S'_{R_t^2} = (S_{R_t^2} \cup \{u\}) \setminus \{v\}$ is a $(P_{R_t^2}, P_{R_t^5})$-DS. Clearly, $S'_{R_t^2} \subseteq P_{R_t^5}$. Consider any $x \in P_{R_t^2} \setminus S'_{R_t^2}$. Since $(OPT \cup \{u\}) \setminus \{v\}$ is a TDS of P, there is $y \in (OPT \cup \{u\}) \setminus \{v\}$ such that $y \cap x \ne \emptyset$. Then $y \in P_{R_t^3}$ by $x \in P_{R_t^2}$, it follows that $y \in S'_{R_t^2}$ by (*). So x is dominated by y under $S'_{R_t^2}$. Thus $S'_{R_t^2}$ is a $(P_{R_t^2}, P_{R_t^5})$-DS.

Then we show that $S'_{R_t^2}$ is total. Since $S_{R_t^2}$ is total, we only need to consider any $z \in S'_{R_t^2}$ intersecting with v. Clearly, $z \in (OPT \cup \{u\}) \setminus \{v\}$. Since $(OPT \cup \{u\}) \setminus \{v\}$ is total, there is $w \in (OPT \cup \{u\}) \setminus \{v\}$ such that $w \cap z \ne \emptyset$. We claim $w \in S'_{R_t^2}$. Note that $w \ne z$. If $w = u$, then $w \in S'_{R_t^2}$. So we assume $w \ne u$, then $w \in OPT \setminus \{v\}$. If $z = u$, then $w \in P_{R_t^3}$ by $u \in P_{R_t^2}$, it follows that $w \in S_{R_t^2} \setminus \{v\} \subseteq S'_{R_t^2}$ by (*). Otherwise $z \ne u$. Since $v \in P_{R_t^3}$, $z \in P_{R_t^4}$ by $v \cap z \ne \emptyset$, further $w \in P_{R_t^5}$ by $w \cap z \ne \emptyset$. Then $w, z \in (OPT \cap P_{R_t^5}) \setminus \{v\}$ and $w \cap z \ne \emptyset$, it implies that w is not isolated in $OPT \cap P_{R_t^5}$. Thus $w \in S_{R_t^2} \setminus \{v\} \subseteq S'_{R_t^2}$. By claim, $S'_{R_t^2}$ is total. Hence $S_{R_t^2}$ is a $(P_{R_t^2}, P_{R_t^5})$-STDS. □

Now we show that the 2-level shifting algorithm is a PTAS.

Lemma 4.4. *$|S| \le (1 + \frac{6}{k})|OPT|$, where OPT is a minimum STDS of P.*

Proof. Consider any (i, j)-th iteration, where $1 \le i, j \le k$. Suppose the lower left corner of $\mathcal{R}$ is (0,0). Let $x = 10i, x = 10i + 10k, \cdots, x = 10i + 10ka_i$ be the vertical lines and $y = 10j, y = 10j + 10k, \cdots, y = 10j + 10kb_j$ be the horizontal lines used in the (i, j)-th partition expect the boundaries of $\mathcal{R}$ for some nonnegative integers a_i and b_j. Let OPT_i^a (resp. OPT_j^b) be the set of squares in OPT whose centers are contained in the strip $[x = 10i + 10ka - 5, x = 10i + 10ka + 5)$ for any $0 \le a \le a_i$ (resp. $[y = 10j + 10kb - 5, y = 10j + 10kb + 5)$ for any $0 \le b \le b_j$). Let $OPT_i^V = \bigcup_{a=0}^{a_i} OPT_i^a$, $OPT_j^H = \bigcup_{b=0}^{b_j} OPT_j^b$ and $OPT_{i,j} = OPT_i^V \cup OPT_j^H$. Note that for any $1 \le i \ne i' \le k$, a square cannot be in both OPT_i^V and $OPT_{i'}^V$, that is, $OPT_i^V \cap OPT_{i'}^V = \emptyset$. Then

$$\sum_{i=1}^{k}|OPT_i^V| = |\bigcup_{i=1}^{k} OPT_i^V| \le |OPT|. \tag{4.1}$$

Similarly, we have

$$\sum_{j=1}^{k}|OPT_j^H| = |\bigcup_{j=1}^{k} OPT_j^H| \le |OPT|. \tag{4.2}$$

By inequalities (4.1) and (4.2), we have

$$\sum_{i=1}^{k}\sum_{j=1}^{k}|OPT_{i,j}| \le k\sum_{i=1}^{k}|OPT_i^V| + k\sum_{j=1}^{k}|OPT_j^H| \le 2k|OPT|. \tag{4.3}$$

For any $1 \le t \le s$, $OPT_{R_t^2}$ is a minimum $(P_{R_t^2}, P_{R_t^5})$-STDS. By Lemma 4.3, $(OPT \cap P_{R_t^5}) \setminus I_{R_t^5}$ is a $(P_{R_t^2}, P_{R_t^5})$-STDS, then

$$|OPT_{R_t^2}| \le |(OPT \cap P_{R_t^5}) \setminus I_{R_t^5}| \le |OPT \cap P_{R_t^5}|. \tag{4.4}$$

Thus

$$\begin{aligned} |S_{i,j}| &\le |OPT_{R_1^2}| + \cdots + |OPT_{R_s^2}| \\ &\le |OPT \cap P_{R_1^5}| + \cdots + |OPT \cap P_{R_s^5}| \\ &\le |OPT| + 3|OPT_{i,j}|. \end{aligned} \tag{4.5}$$

The second inequality is obtained by (4.4). The third inequality is obtained by the fact that only the squares in $OPT_{i,j}$ can be repetitively calculated and each square is repetitively calculated at most 4 times.

Therefore by inequalities (4.3) and (4.5) we have

$$\begin{aligned} |S| &= \min_{1\le i,j\le k}|S_{i,j}| \\ &\le \frac{1}{k^2}\sum_{i=1}^{k}\sum_{j=1}^{k}|S_{i,j}| \\ &\le \frac{1}{k^2}(k^2|OPT| + 3\sum_{i=1}^{k}\sum_{j=1}^{k}|OPT_{i,j}|) \\ &\le \frac{1}{k^2}(k^2|OPT| + 6k|OPT|) \\ &= (1+\frac{6}{k})|OPT|. \quad \square \end{aligned}$$

Theorem 4.5. *Given a set P of n unit squares such that the corresponding unit square graph has no isolated vertex and an integer $k \ge 1$. The proposed 2-level shifting algorithm produces an STDS of P with size at most $(1+\frac{6}{k})|OPT|$ in time $n^{O(k^2)}$, where OPT is a minimum STDS of P.*

Proof. By Lemma 4.2 and Lemma 4.4, the algorithm produces an STDS of P with size at most $(1+\frac{6}{k})|OPT|$. Now we discuss the running time. For each (i, j)-th iteration $(1 \le i, j \le k)$, the time taken to compute OPT_{R^2} is $n_{R^5}^{O(k^2)}$ by Lemma 4.1, where R is any one of the rectangles in this iteration and $n_{R^5} = |P_{R^5}|$. Since each square participates in computing $S_{i,j}$ at most 4 rectangles, computing $S_{i,j}$ needs $n^{O(k^2)}$ time. There are k^2 iterations, the total running time is $n^{O(k^2)}$. $\square$

When P is a set of unit disks, the 2-level shifting algorithm is also effective. Hence we have the following result.

Corollary 4.6. *Given a set P of n unit disks such that the corresponding unit disk graph has no isolated vertex and an integer $k \ge 1$. The proposed 2-level shifting algorithm produces an STDS of P with size at most $(1+\frac{6}{k})|OPT|$ in time $n^{O(k^2)}$, where OPT is a minimum STDS of P.*

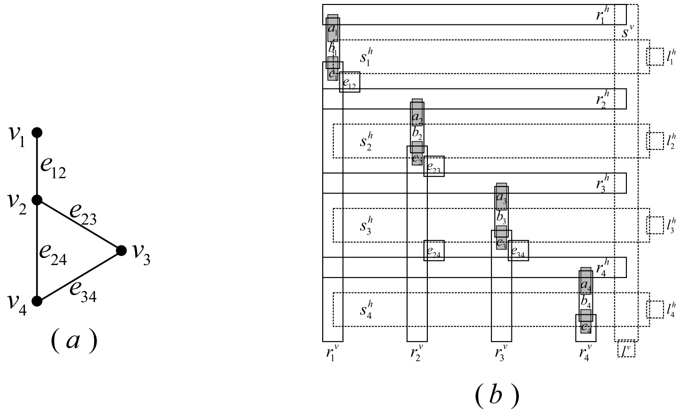

Fig. 16. (a) A graph G of degree at most 3. (b) A rectangle intersection representation of a proper rectangle graph G' obtained from G. Rectangles drawn with dashed lines represent rectangles in SL.

5. APX-hardness

Note that unit square graphs and unit-height rectangle graphs are subclasses of rectangle graphs. In section 3 and section 4, we give some approximation algorithms for the MSCDS problem and the MSTDS problem in unit square graphs and unit-height rectangle graphs. In this section, we consider the inapproximability of the MSCDS problem and the MSTDS problem in rectangle graphs. We show that the two problems are APX-hard even in proper rectangle graphs which are a subclass of rectangle graphs. Further we give an explicit lower bound 1.00147 on efficient approximability for the two problems in proper rectangle graphs unless P=NP.

We denote the Min-3-VC problem by a problem of finding a minimum vertex cover in graphs of degree at most 3.

Theorem 5.1. *The MSCDS problem and the MSTDS problem in proper rectangle graphs are APX-hard.*

Proof. We give an L-reduction from the APX-hard problem the Min-3-VC problem [3] to the MSCDS problem and the MSTDS problem in proper rectangle graphs. Let $G = (\{v_1, \cdots, v_n\}, E)$ be an instance of the Min-3-VC problem and $|E| = m$. For each $v_iv_j \in E$, we assume $1 \le i < j \le n$, denote the edge by e_{ij}, that is, $e_{ij} = v_iv_j$. Using the similar method as in [14], we construct a proper rectangle graph G' as follows:
Step 1. For each vertex v_i, construct 5 rectangles $r_i^v, r_i^h, a_i, b_i, c_i$. In details, $r_1^h, \cdots, r_n^h$ are same thin horizontal rectangles and are placed from top to bottom without intersecting each other such that their left boundaries are on a vertical line l. For each $1 \le i \le n$, r_i^v is a thin vertical rectangle. $r_1^v, \cdots, r_n^v$ are placed from left to right without intersecting each other such that: (1) the left boundary of r_1^v is on the vertical line l, (2) the top boundary of r_i^v is between the bottom boundary of r_i^h and top boundary of r_{i+1}^h for each $1 \le i \le n$ (r_{n+1}^h can be regarded as a virtual rectangle below the r_n^h). a_i, b_i, c_i are placed such that: (1) b_i intersects r_i^v, r_i^h, (2) a_i (resp. c_i) intersects b_i, r_i^h (resp. b_i, r_i^v). By the above structure, there is a region s_{ij} where r_i^v and r_j^h intersect for any $1 \le i < j \le n$.
Step 2. For each $e_{ij} \in E$, construct a little square e_{ij} such that e_{ij} intersects r_i^v, r_j^h. This can be achieved by placing e_{ij} intersects the region s_{ij}.
Step 3. Construct a thin vertical rectangle s^v intersecting with the right boundary of r_i^h for each $1 \le i \le n$ and a little square l^v only intersecting with s^v. For each $1 \le i \le n$, construct a horizontal rectangle s_i^h between the bottom boundary of r_i^h and the top boundary of r_{i+1}^h such that s_i^h is intersecting with $s^v, a_i, b_i, c_i, r_1^v, \cdots, r_i^v$ and $e_{t(i+1)}$ (if $e_{t(i+1)}$ exists), where $1 \le t \le i$. Then construct a little square l_i^h only intersecting with s_i^h. Let $LS = \{s^v, l^v, s_i^h, l_i^h : 1 \le i \le n\}$.
Step 4. Appropriately adjust the size and position of the rectangles will make no rectangle is properly contained within another. Fig. 16 (b) is a rectangle intersection representation of a proper rectangle graph G' obtained from G given in Fig. 16 (a).

Let C be a minimum vertex cover of G. Construct a set S from C by adding r_i^h and r_i^v to S for each $v_i \in C$, adding b_i to S for each $v_i \notin C$ and adding the set LS to S. Clearly, LS is a CDS of G' and $S \setminus LS$ is a DS of $G' \setminus LS$ by C is a vertex cover of G. Thus by Lemma 1.3, S is an SCDS of G', also an STDS of G'. Since the degree of G is at most 3, $\beta(G) \ge n/4$. Hence $\gamma_{st}(G') \le \gamma_{sc}(G') \le |S| = |C| + 3n + 2 = \beta(G) + 3n + 2 \le 14\beta(G)$ (note that we can assume that the graph G is not a star, then $\beta(G) \ge 2$).

Let S be an SCDS or STDS of G'. By Theorem 1.1, $LS \subseteq S$ and each vertex in LS is not an S-defender. For each $x \notin S$, there is an S-defender $y \in S \setminus LS$ of x. Thus $D = S \setminus LS$ is a DS of $G' \setminus LS$. We will construct a DS D' of $G' \setminus LS$ from D satisfying the following conditions: (1) $|D'| \le |D|$; (2) for each $e_{ij} \in E$, $e_{ij} \notin D'$; (3) $D' \cap H_i = \{r_i^h, r_i^v\}$ or $D' \cap H_i = \{b_i\}$ for each $1 \le i \le n$, where $H_i = \{r_i^h, r_i^v, a_i, b_i, c_i\}$. For each $e_{ij} \in D$, if $r_i^v \in D$ or $r_j^h \in D$, then let $D_1 = D \setminus \{e_{ij}\}$. Otherwise $r_i^v, r_j^h \notin D$, then let

$D_1 = (S_1 \setminus \{e_{ij}\}) \cup \{r_j^h\}$. In both cases, D_1 is a DS of $G' \setminus LS$ with $|D_1| \le |D|$ and $e_{ij} \notin D_1$ for any $e_{ij} \in E$. For each $1 \le i \le n$, since a_i, b_i, c_i need to be dominated by D_1, $|D_1 \cap H_i| \ge 1$. If $|D_1 \cap H_i| = 1$, then $D_1 \cap H_i = \{b_i\}$. If $|D_1 \cap H_i| \ge 2$, then let $D' = (D_1 \setminus H_i) \cup \{r_i^h, r_i^v\}$. Obviously, D' is a DS of $G' \setminus LS$ satisfying the conditions (1), (2) and (3). Since LS is a CDS of G', $S' = D' \cup LS$ is an SCDS of G' by Lemma 1.3, also an STDS of G'.

Let $C = \{v_i \in V : r_i^v, r_i^h \in D'\}$. For each $e_{ij} \in E$, $e_{ij} \notin D'$ by (2). Then $r_i^v \in D'$ or $r_j^h \in D'$, without loss of generality, suppose $r_i^v \in D'$. Since a_i needs to be dominated by D', $|D' \cap H_i| \ge 2$. Thus $D' \cap H_i = \{r_i^h, r_i^v\}$ by condition (3), it follows that $v_i \in C$. So C is a vertex cover of G. By conditions (1), (2) and (3), $|S| \ge |S'| = |C| + 3n + 2 \ge \beta(G) + 3n + 2$. If S is a minimum SCDS or STDS of G', then $\gamma_{sc}(G') = |S| \ge \beta(G) + 3n + 2$ or $\gamma_{st}(G') = |S| \ge \beta(G) + 3n + 2$. Hence,

$$\gamma_{st}(G') = \gamma_{sc}(G') = \beta(G) + 3n + 2. \tag{1}$$

Thus, $|S| - \gamma_{sc}(G') \ge |C| - \beta(G)$ and $|S| - \gamma_{st}(G') \ge |C| - \beta(G)$. $\square$

Finally, we give an explicit lower bound on efficient approximability for the MSCDS problem and the MSTDS problem in proper rectangle graphs unless P=NP.

Theorem 5.2. *The MSCDS problem and the MSTDS problem do not have a polynomial-time* 1.00147*-approximation in proper rectangle graphs unless P=NP.*

Proof. We use the NP-hard gap for the MIN-3-VC problem [11]. For any $\varepsilon > 0$, it is NP-hard to decide in a graph $G = (V, E)$ of degree at most 3 whether $\beta(G) < \frac{|V|}{2}(1 + 2\delta_3 + \varepsilon)$ or $\beta(G) > \frac{|V|}{2}(1 + 3\delta_3 - \varepsilon)$, where $\delta_3 \approx 0.0103305$. Let G' be a proper rectangle graph obtained from G as Theorem 5.1. By the formula (1) $\gamma_{st}(G') = \beta(G) + 3|V| + 2$, we obtain it is NP-hard to decide whether $\gamma_{st}(G') < \frac{|V|}{2}(1 + 2\delta_3 + \varepsilon) + 3|V| + 2$ or $\gamma_{st}(G') > \frac{|V|}{2}(1 + 3\delta_3 - \varepsilon) + 3|V| + 2$ for any $\varepsilon > 0$. Hence the MSTDS problem does not have a polynomial-time 1.00147-approximation in proper rectangle graphs unless P=NP. Similarly, the MSCDS problem does not have a polynomial-time 1.00147-approximation in proper rectangle graphs unless P=NP. $\square$

6. Conclusion

In this paper, we studied complexity and algorithmic aspects of the MSCDS problem and the MSTDS problem in unit disk graphs and rectangle graphs. In section 2, we show that the decision version of the MSCDS problem is NP-complete in unit square graphs and unit disk graphs. Then we show that the decision version of the MSTDS problem is NP-complete even in grid graphs. In section 3, we give linear-time constant-approximation algorithms for the two problems in unit square graphs, unit disk graphs and unit-height rectangle graphs. In section 4, we propose a PTAS for the MSTDS problem in unit square graphs and unit disk graphs. In section 5, we show that the two problems in proper rectangle graphs are APX-hard. Further we give an explicit lower bound 1.00147 on efficient approximability for the two problems in proper rectangle graphs unless P=NP. In the following we give some open problems.

1. Is the decision version of the MSCDS problem NP-complete in grid graphs?
2. Is there a constant-approximation algorithm for the two problems on rectangle graphs?
3. Is there a PTAS for the MSCDS problem in unit disk graphs or unit square graphs?

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work is supported by National Natural Science Foundation of China under Grant 12071194, 11571155.

References

[1] P.K. Agarwal, N.H. Mustafa, Independent set of intersection graphs of convex objects in 2D, Comput. Geom. 34 (2) (2006) 83–95.
[2] P.K. Agarwal, M. van Kreveld, S. Suri, Label placement by maximum independent set in rectangles, Comput. Geom. 11 (3–4) (1998) 209–218.
[3] P. Alimonti, V. Kann, Some APX-completeness results for cubic graphs, Theor. Comput. Sci. 237 (1–2) (2000) 123–134.
[4] S. Benecke, E.J. Cockayne, C.M. Mynhardt, Secure total domination in graphs, Util. Math. 74 (2007) 247–259.
[5] A.A. Bertossi, M.A. Bonuccelli, Hamiltonian circuits in interval graph generalizations, Inf. Process. Lett. 23 (4) (1986) 195–200.
[6] T. Biedl, G. Kant, A better heuristic for orthogonal graph drawings, Comput. Geom. 9 (3) (1998) 159–180.

[7] A.G. Cabaro, S.R. Canoy Jr., Secure connected dominating sets in the join and composition of graphs, Int. J. Math. Anal. 9 (25) (2015) 1241–1248.
[8] A.G. Cabaro, S.S. Canoy Jr., I.S. Aniversario, Secure connected domination in a graph, Int. J. Math. Anal. 8 (42) (2014) 2065–2074.
[9] T.M. Chan, Polynomial-time approximation schemes for packing and piercing fat objects, J. Algorithms 46 (2) (2003) 178–189.
[10] X. Cheng, X. Huang, D. Li, W. Wu, D.-Z. Du, A polynomial-time approximation scheme for the minimum-connected dominating set in ad hoc wireless networks, Networks 42 (4) (2003) 202–208.
[11] M. Chlebík, J. Chlebíková, Complexity of approximating bounded variants of optimization problems, Theor. Comput. Sci. 354 (3) (2006) 320–338.
[12] E.J. Cockayne, P.J.P. Grobler, W.R. Gründlingh, J. Munganga, J.H. van Vuuren, Protection of a graph, Util. Math. 67 (2005) 19–32.
[13] O. Duginov, Secure total domination in graphs: bounds and complexity, Discrete Appl. Math. 222 (2017) 97–108.
[14] T. Erlebach, E.J. van Leeuwen, Domination in geometric intersection graphs, in: LATIN, Springer, 2008, pp. 747–758.
[15] S. Funke, A. Kesselman, U. Meyer, M. Segal, A simple improved distributed algorithm for minimum CDS in unit disk graphs, ACM Trans. Sens. Netw. 2 (3) (2006) 444–453.
[16] W.K. Hale, Frequency assignment: theory and applications, Proc. IEEE 68 (12) (1980) 1497–1514.
[17] J. Hartmanis, Computers and intractability: a guide to the theory of NP-completeness (Michael R. Garey and David S. Johnson), SIAM Rev. 24 (1) (1982) 90–91.
[18] D.S. Hochbaum, W. Maass, Approximation schemes for covering and packing problems in image processing and VLSI, J. ACM 32 (1) (1985) 130–136.
[19] R.K. Jallu, P.R. Prasad, G.K. Das, Distributed construction of connected dominating set in unit disk graphs, J. Parallel Distrib. Comput. 104 (2017) 159–166.
[20] S.K. Jena, G.K. Das, Total domination in geometric unit disk graphs, in: CCCG, 2021.
[21] A. Jha, Secure total domination in chain graphs and cographs, AKCE Int. J. Graphs Comb. 17 (3) (2020) 826–832.
[22] J.P. Kumar, P.V.S. Reddy, Algorithmic aspects of secure connected domination in graphs, Discuss. Math., Graph Theory 41 (4) (2021) 1179–1197.
[23] J.P. Kumar, P.V.S. Reddy, S. Arumugam, Algorithmic complexity of secure connected domination in graphs, AKCE Int. J. Graphs Comb. 17 (3) (2020) 1010–1013.
[24] M. Li, P.-J. Wan, F. Yao, Tighter approximation bounds for minimum CDS in wireless ad hoc networks, in: Proc. ISAAC, vol. 5878, 2009, pp. 699–709.
[25] A.C. Martínez, A. E.-Moreno, J.A. R.-Velázquez, Secure total domination in rooted product graphs, Mathematics 8 (2020) 1–14.
[26] A. Poureidi, Algorithmic results in secure total dominating sets on graphs, Theor. Comput. Sci. 918 (2022) 1–17.
[27] L.G. Valiant, Universality considerations in VLSI circuits, IEEE Trans. Comput. 30 (1981) 135–140.
[28] P.-J. Wan, K.M. Alzoubi, O. Frieder, Distributed construction of connected dominating set in wireless ad hoc networks, Mob. Netw. Appl. 9 (2004) 141–149.
[29] C.-X. Wang, Y. Yang, S.-J. Xu, Algorithmic aspects of secure domination in unit disk graphs, 2022, manuscript.
[30] C.-X. Wang, Y. Yang, S.-J. Xu, Algorithmic aspects of secure domination in rectangle intersection graphs, 2022, manuscript.
[31] Y. Yang, C.-X. Wang, S.-J. Xu, Total (restrained) domination in unit disk graphs, 2022, manuscript.